

Carve Engineering Spec

A *living* document describing the **Carve** pipeline. See the [spec](#) and `Carve.parse` for the AST.

Architecture

The parser runs in **two passes**:

1. Block identification (whole document)
2. Inline parsing (delimiter stack, no backtracking)
 - links and images
 - emphasis: **strong**, *emphasis*, ***bold italic***
 - `inline code`

Design principle: plain text should render beautifully and stay readable at the source.

Carve design doc

Engines

Engine	Language	Status
carve-js	TypeScript	reference
carve-rs	Rust	reference
carve-php	PHP	full
carve-rb	Ruby (FFI)	full

Task status

- ✓ Block parser
- ✓ Inline parser
- ✓ HTML renderer
- PDF renderer polish
 - ✓ tables
 - ✓ math images

Notes

Term

A single unit of markup meaning.

AST

The parsed tree returned by `Carve.parse`.

```
pub fn parse(source: &str) -> Document {  
    // block pass, then inline pass  
    Document::from(source)  
}
```

Printed through Chrome DevTools by `crv2pdf` , from a faithful Carve HTML fragment.

By Mark Scherer · 2026-07-20